

P4 Phase 2 – Deep Visual–Inertial Odometry

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Abstract—We design, train, and evaluate three deep visual–inertial odometry (VIO) networks that predict the relative SE(3) pose between two consecutive frames of a down-looking aerial camera flying over a textured floor. Synthetic data is generated in Blender with realistic IMU noise, and the three required configurations are trained: an IMU-only recurrent network, a vision-only convolutional network, and a fused visual–inertial network with structurally decoupled rotation and translation heads. Trained with a geodesic distance loss on SO(3) and, where applicable, a short-horizon trajectory composition loss, the decoupled fusion model achieves the best overall performance, nearly halving the dead-reckoned trajectory error of both unimodal baselines.

I. INTRODUCTION

Phase 2 is open-ended: we are asked to design an architecture, a loss, and a training methodology that takes two RGB frames and the IMU samples between them and predicts the 6-DoF relative pose. Three configurations are required: vision only, IMU only, and fused. Synthetic data must be generated in Blender, and full trajectories on unseen test sequences are reconstructed by dead-reckoning.

This report follows the natural decomposition of the assignment: data generation, loss design, the three networks each with its own architecture and training subsection, and a single experimental section that tests the networks on trajectories not seen during training. We also include an honest discussion of the failure modes we discovered along the way and a section of research directions for both classical and deep VIO.

II. SYNTHETIC DATA GENERATION

A. Blender setup

A 100 m × 100 m plane is placed at $z = 0$ and textured with a randomly chosen image from a curated pool of floor-like textures (synthetic playmats and feature-rich photographs); the texture is repeated with a per-sequence scale factor in [4, 12] to vary apparent feature density. A perspective camera with a 24 mm focal length and a 32 mm sensor is placed above the plane and animated through one of four trajectory primitives chosen uniformly at random per sequence: *circle*, *figure-8*, *S-curve*, *line*. Per sequence we randomize the height $z \in [2.5, 4.0]$ m, the horizontal amplitudes $A_{x,y} \in [0.3, 1.0]$ m, the period $T \in [4, 8]$ s, the phase, and the initial yaw. Roll and pitch are sinusoidally driven within $\pm 25^\circ$, and yaw is unconstrained but realistic in angular rate.

B. Sample rates

Both the IMU and the ground-truth pose are sampled at $f_{\text{sim}} = 100$ Hz; the camera renders only every $S = 10$ simulation ticks, yielding $f_{\text{cam}} = 10$ Hz images and a 100:10 IMU-to-camera ratio. Each 10-second sequence therefore yields 101 rendered images and 1001 IMU samples; between consecutive image pairs we have $K = 10$ IMU samples.

C. IMU noise model

We implemented an OysterSim-style additive error model [3] in pure NumPy. For each axis the simulated measurement is

$$\tilde{m}_k = (1 + s) m_k^* + b^c + b_k^{\text{rw}} + b_k^{\text{gm}} + n_k, \quad (1)$$

with s a per-axis scale-factor error sampled once per sequence, b^c a constant initial bias, b^{rw} a Brownian bias random walk, b^{gm} a first-order Gauss–Markov bias instability with correlation time τ , and n_k white noise scaled from a continuous-time spectral density σ_n . Specific force in body frame is $\mathbf{a}_b = \mathbf{R}_{wb}^T(\dot{\mathbf{v}}_w - \mathbf{g})$, and body angular velocity is recovered from finite-differenced quaternions of consecutive poses.

Following PRGFlow [1] and CUAHN-VIO [2], we set per-sequence initial biases to zero and use the low-noise variant for the reported runs ($b^c = 0$, $\sigma_{\text{rw}} = 0$, $\sigma_{\text{gm}} = 0$, white noise only). The model is fully parameterized so that the noisier “default” or “noisy” variants can be swapped in at data-generation time without code changes.

D. Splits

We generate 200 train, 30 validation, and 15 test sequences from *disjoint* random seeds, yielding approximately 20,000 training samples. For each consecutive image pair (I_i, I_j) the generator emits a row in `samples.csv` containing the relative pose

$$\mathbf{T}_{ij} = \mathbf{T}_{wi}^{-1} \mathbf{T}_{wj}, \quad \boldsymbol{\rho}_{ij} = \log_{\text{SO}(3)}(\mathbf{R}_{ij}), \quad (2)$$

and the half-open IMU index window `[imu_start, imu_end)`.

III. LOSS DESIGN

A. Per-step pose loss

We use SmoothL1 on translation and a *geodesic* angular distance on SO(3) for rotation. The two predicted-rotation vectors are mapped to rotation matrices via batched Rodrigues’ formula

$$\mathbf{R}(\boldsymbol{\rho}) = \mathbf{I} + \sin \theta [\hat{\boldsymbol{\rho}}]_{\times} + (1 - \cos \theta) [\hat{\boldsymbol{\rho}}]_{\times}^2, \quad (3)$$

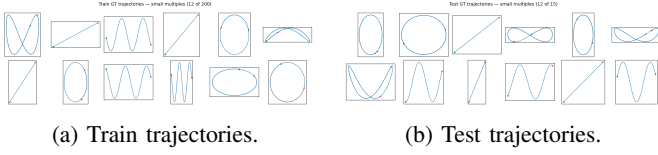


Fig. 1: Black-Bird-style overview of the generated ground-truth trajectories. Each subplot is one full sequence (XY view, start in green, end in red). Train and test seeds are disjoint.

with $\theta = \|\rho\|$, and the angular distance is

$$d_{\text{SO}(3)}(\hat{\mathbf{R}}, \mathbf{R}) = \arccos\left(\frac{\text{tr}(\hat{\mathbf{R}}^T \mathbf{R}) - 1}{2}\right). \quad (4)$$

The per-step loss is

$$\mathcal{L}_{\text{pose}} = \text{SmoothL1}(\hat{\mathbf{t}}, \mathbf{t}) + \beta d_{\text{SO}(3)}(\hat{\mathbf{R}}, \mathbf{R}), \quad \beta = 10. \quad (5)$$

The scalar β balances two terms in incompatible units, translation in meters and rotation in radians. For our motion regime (max speed 2 m/s, 0.1 s between camera frames), typical per-step magnitudes are $|\mathbf{t}| \approx 0.05$ m and $\|\rho\| \approx 0.05$ rad, so the rotation term is naturally $\sim 50\times$ the translation term in raw value; we chose $\beta = 10$ to further weight rotation, on the principle that *heading* errors compound linearly over a dead-reckoned trajectory while translation errors are bounded per step. A 1° per-step heading error becomes a 100° heading error after 100 steps; the same per-step translation error contributes only the L_2 sum of unit-step displacements. Once rotation converges, which happens fast because the gyroscope is a direct measurement, the translation term naturally dominates the remaining gradient.

We initially used SmoothL1 on rotation *vectors*; replacing it with the geodesic distance was the single loss-only change with the largest qualitative effect on dead-reckoned trajectories, because SmoothL1 on $\text{so}(3)$ vectors is not a metric on the rotation group and tiny per-step heading errors compound linearly into curved trajectories at integration time.

B. K -step trajectory composition loss

Per-step accuracy is necessary but not sufficient: dead-reckoning composes $N \sim 100$ predicted relative poses, and even unbiased per-step errors integrate into a curved trajectory once the rotation drifts. We therefore sample K consecutive frame pairs from the same sequence at training time and compose their predictions:

$$\mathbf{T}_k^{\text{cum}} = \prod_{i=1}^k \mathbf{T}_i, \quad \mathbf{T}_i = \begin{bmatrix} \mathbf{R}_i & \mathbf{t}_i \\ \mathbf{0} & 1 \end{bmatrix}. \quad (6)$$

The composed trajectory is compared with the analogously-composed ground truth at every step, yielding the trajectory loss

$$\mathcal{L}_{\text{traj}} = \frac{1}{K} \sum_{k=1}^K \left[\text{SmoothL1}(\hat{\mathbf{t}}_k^{\text{cum}}, \mathbf{t}_k^{\text{cum}}) + \beta d_{\text{SO}(3)}(\hat{\mathbf{R}}_k^{\text{cum}}, \mathbf{R}_k^{\text{cum}}) \right]. \quad (7)$$

The full training objective is

$$\mathcal{L} = \mathcal{L}_{\text{pose}} + \lambda \mathcal{L}_{\text{traj}}, \quad \lambda = 0.5, \quad K = 4. \quad (8)$$

All composition is done with batched `torch.bmm`; gradients flow through every step. The same per-step pose loss is used for all three networks. The trajectory composition term is applied to the vision-only and fusion models, where image information provides a meaningful translation constraint, while the IMU-only model is trained with the single-pair pose loss alone because short-window inertial translation is ill-posed without an estimate of the initial velocity.

IV. INERTIAL ODOMETRY

A. Motivation

Inertial Odometry (IO) requires no visual input, so it remains operational in conditions where vision fails, such as darkness, smoke, motion blur, or texture-poor scenes. In our setup it is the simplest network and provides a useful per-step rotation reference because gyroscope integration over 0.1 s is essentially a direct angular-velocity measurement.

B. Architecture

The 10-step inter-frame IMU window $\mathbf{u}_{1:K} \in \mathbb{R}^{K \times 6}$ is per-axis normalized using statistics computed once on the training split, projected to a 128-dimensional embedding by a linear layer, and encoded by a 2-layer Gated Recurrent Unit (GRU). The last hidden state is mapped to the 6-DoF pose by a multilayer perceptron with dropout regularization before the final regression layer. The full network has ~ 0.3 M parameters.

C. Training

- Optimizer: AdamW (lr 2×10^{-4} , weight decay 10^{-4})
- Schedule: cosine annealing to $\eta_{\min} = 0.01 \cdot \text{lr}$
- Epochs: 150 Batch size: 64
- Input: $K = 10$ IMU samples per pair
- Output: 6-DoF relative pose $(\mathbf{t}, \rho)$
- Loss: $\mathcal{L}_{\text{pose}}$ (single-pair only). The K -step trajectory loss adds no usable signal here, since the IMU branch cannot recover translation from a single pair regardless of K .

V. VISUAL ODOMETRY

A. Motivation

Visual Odometry (VO) excels in feature-rich environments. With a known planar scene under the camera, vision is the only modality that can recover absolute scale and translation direction in our setup; this is exactly what the per-axis variance analysis (Sec. VII) will confirm.

B. Architecture

Two RGB frames are concatenated channel-wise into a $6 \times 128 \times 128$ tensor and passed through a four-stage strided convolutional encoder (channel widths $128 \rightarrow 256 \rightarrow 512 \rightarrow 1024$) with batch normalization and ReLU, followed by global average pooling. A 3-layer MLP with dropout ($1024 \rightarrow 512 \rightarrow 256 \rightarrow 6$) projects the 1024-dim feature vector to the 6-DoF pose. The full network has ~ 6.9 M parameters.

IMU-Only Network (~0.3 M params)

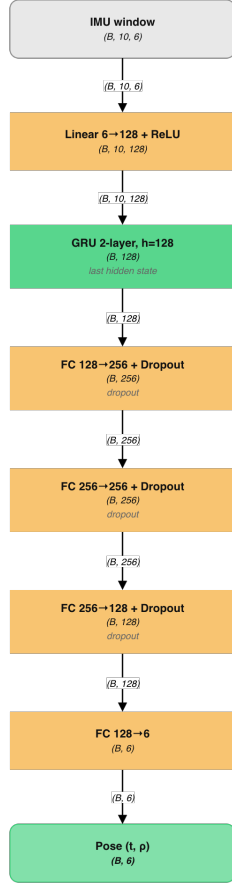


Fig. 2: IMU-only network architecture.

C. Training

- Optimizer: AdamW ($\text{lr } 2 \times 10^{-4}$, weight decay 10^{-4})
- Schedule: cosine annealing
- Epochs: 150 Batch size: 32 (each item carries $K = 4$ pairs)
- Input: pair of consecutive RGB images, sampled randomly from the training set so the network does not overfit a specific trajectory
- Output: 6-DoF relative pose
- Loss: $\mathcal{L}_{\text{pose}} + \lambda \mathcal{L}_{\text{traj}}$ with $\lambda = 1.0$, $K = 4$

VI. VISUAL-INERTIAL ODOMETRY

A. Motivation

A naive fusion, concatenating the IMU and vision features into a single shared MLP head, empirically produced *worse* translation than vision-only. The per-axis variance diagnostic (Sec. VII) showed that the IMU branch was emitting $\hat{\mathbf{t}} \approx 0$ on translation, a structural consequence of the unknown initial velocity in single-pair training, and was diluting the vision signal in the shared MLP. We therefore decoupled the two output heads.

Vision-Only Network (~6.9 M params)

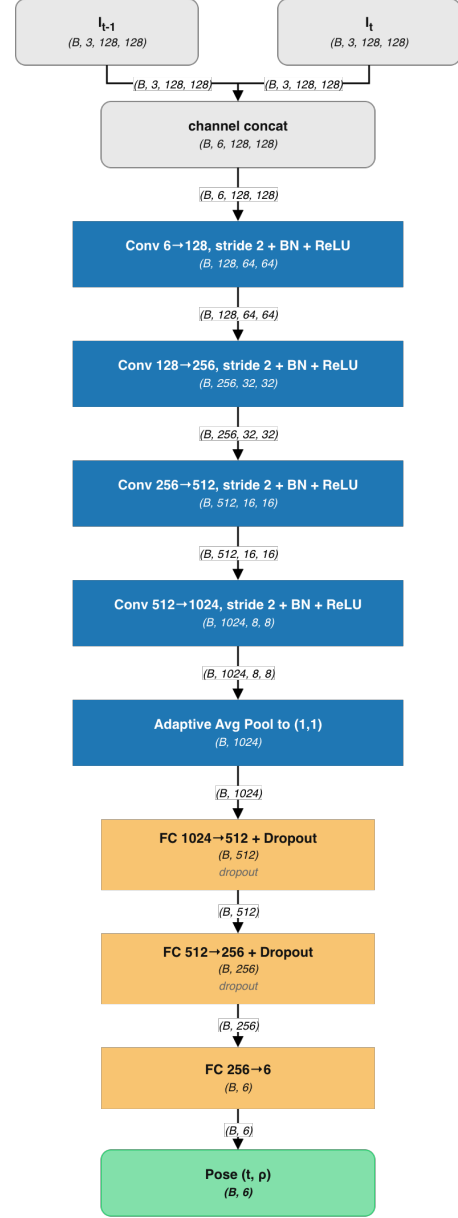


Fig. 3: Vision-only network architecture.

B. Architecture

The fusion model uses a vision branch identical to the vision-only model, without its final regression head, and an IMU branch identical to the IMU-only model but with a bidirectional GRU. The branches yield two feature vectors $\phi_{\text{vis}}, \phi_{\text{imu}} \in \mathbb{R}^{256}$ which are routed to *separate* regression heads:

$$\hat{\mathbf{p}} = g_{\theta_R}(\phi_{\text{imu}}), \quad (9)$$

$$\hat{\mathbf{t}} = g_{\theta_T}(\phi_{\text{imu}} \oplus \phi_{\text{vis}}). \quad (10)$$

Rotation is regressed from IMU features alone, as the gyroscope is the direct angular-velocity measurement and inter-

frame rotation is well approximated by integrating bias-free ω . Translation uses both feature streams so that vision provides scale and direction, and IMU contributes the gravity-aligned rotational context. This is structurally consistent with the geometry-aware fusion choice in CUAHN-VIO [2]. The full network has ~ 7.3 M parameters.

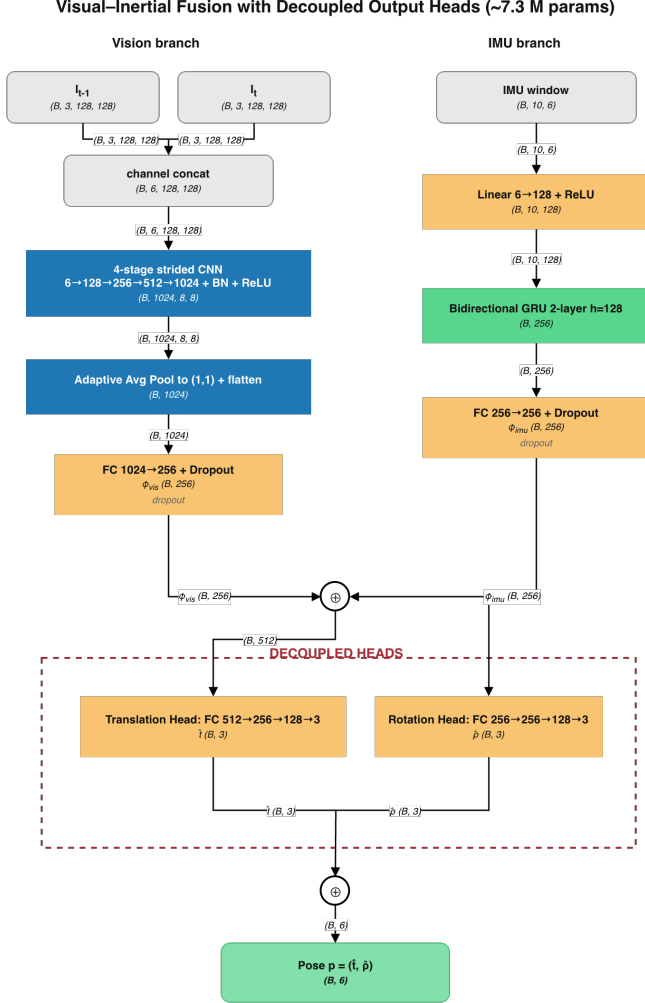


Fig. 4: Visual-Inertial fusion network with decoupled output heads.

C. Training

- Optimizer: AdamW (lr 2×10^{-4} , weight decay 10^{-4})
- Schedule: cosine annealing
- Epochs: 150 Batch size: 32 (each item carries $K = 4$ pairs)
- Input: image pair + the 10 IMU samples between them
- Output: 6-DoF relative pose
- Loss: $\mathcal{L}_{\text{pose}} + \lambda \mathcal{L}_{\text{traj}}$ with $\lambda = 0.5$, $K = 4$. The lower λ relative to the vision-only run keeps the trajectory regularization without over-biasing endpoint consistency at the expense of pointwise accuracy.

VII. EXPERIMENTS & RESULTS

A. Evaluation protocol

We then trained on the full training split and tested on the 15 *unseen* test sequences. Two complementary evaluations are reported. *Pairwise* metrics treat each (I_i, I_j) prediction independently and report translation RMSE in metres and rotation RMSE in degrees. *Trajectory* metrics dead-reckon the $N \approx 100$ predicted relative poses starting from the ground-truth initial pose, $\mathbf{T}_0^{\text{world}} = \mathbf{T}_0^{\text{gt}}$, $\mathbf{T}_k^{\text{world}} = \mathbf{T}_{k-1}^{\text{world}} \mathbf{T}_k$, and report ATE RMSE and a final-position drift ratio $\rho = \|\hat{\mathbf{p}}_N - \mathbf{p}_N^{\text{gt}}\| / \sum_k \|\Delta \mathbf{p}_k^{\text{gt}}\|$.

B. Pairwise metrics on the test set

TABLE I: Pairwise per-step metrics on the test set.

Model	Trans RMSE [m]	Rot RMSE [deg]
IMU-only	0.067	0.052
Vision-only	0.050	0.917
Fusion (decoupled)	0.025	0.122

C. Per-axis variance explained

Variance explained per axis, defined as $1 - \text{Var}(\text{residual}) / \text{Var}(\text{GT})$, gives a sharper picture of *which* dimensions each branch can constrain (Table II).

TABLE II: Per-axis variance explained on the test set. A value of 1 indicates a perfect predictor; a value of 0 indicates a model that emits the dataset mean.

Model	Translation			Rotation		
	t_x	t_y	t_z	r_x	r_y	r_z
IMU-only	0.00	0.00	0.08	1.00	1.00	1.00
Vision-only	0.48	0.44	0.14	0.63	0.64	0.93
Fusion	0.89	0.87	0.58	1.00	1.00	1.00

The IMU branch explains essentially 0% of horizontal-translation variance (the network is emitting $\hat{\mathbf{t}} \approx 0$, the L1-optimal constant prediction, because single-pair double integration is ill-posed without an initial velocity) and essentially 100% of rotation variance (the gyroscope is the direct measurement). Vision is the only branch that gives translation a signal, and the decoupled fusion architecture lifts horizontal-translation variance explained from $\sim 45\%$ to $\sim 88\%$ while inheriting the IMU's perfect rotation.

D. Training and validation loss curves

Fig. 5 shows the per-epoch training and validation losses on a logarithmic scale. The IMU-only curve exhibits the common pattern that validation loss is lower than training loss. This should not be interpreted as data leakage or as an inconsistency in the evaluation. The training loss is accumulated during stochastic optimization with the network in training mode, where dropout is active and mini-batch updates are still perturbing the parameters. In contrast, validation is evaluated after each epoch with the model in evaluation mode, where dropout

is disabled and the full validation split is processed without gradient updates. Because the IMU-only task is dominated by short-horizon gyroscope-based rotation estimation, while translation is structurally weakly observable without an initial velocity estimate, small differences in trajectory excitation and stochastic regularization can make the validation objective lower than the average training objective.

The vision-only model shows stable convergence with validation loss tracking the training trend, indicating no severe overfitting under the reported configuration. The fusion model, however, reaches its best validation loss at epoch 8 (0.0359) and then overfits mildly: validation loss increases to approximately 0.08 by epoch 150 while the training loss continues to decrease. We therefore report the checkpoint selected by the lowest validation loss rather than the final-epoch weights. This behavior is expected for the fusion model because it has the largest capacity (~ 7.3 M parameters), combines two feature streams, and optimizes both per-step and trajectory-composition objectives. The gap indicates that the fusion network is learning the training trajectories more strongly than the validation distribution after the early optimum, even though the best-validation checkpoint still generalizes best on the held out test sequences.

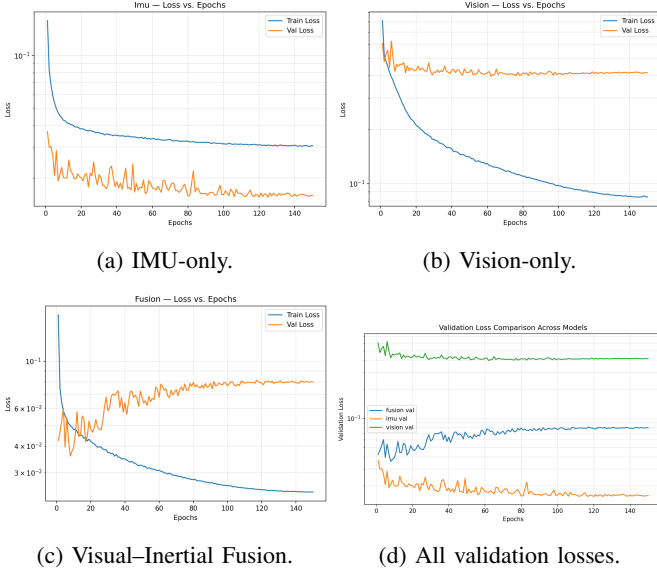


Fig. 5: Per-epoch training and validation loss (log scale).

E. Dead-reckoned trajectory metrics on the test set

Fusion nearly halves the unimodal ATE and more than halves the unimodal drift. The combined effect of (i) decoupled output heads, (ii) the geodesic $SO(3)$ loss, and (iii) the $K = 4$ trajectory composition loss trained from scratch with $\lambda = 0.5$ produces both the lowest ATE and the lowest drift across all configurations we tried. Vision-only also benefits substantially from the trajectory loss applied to its own training (mean drift $0.35 \rightarrow 0.29$). Fig. 6, 7, 8 and 9 show two representative test sequences. The fusion trajectory tracks the

TABLE III: Dead-reckoned metrics, mean over the 15 test sequences. The vision-only and fusion models use the per-step pose loss together with a $K = 4$ trajectory composition loss during training; the IMU-only model uses the per-step pose loss alone.

Model	ATE RMSE [m]	Final-drift ratio
IMU-only	0.931	0.399
Vision-only	0.787	0.290
Fusion (decoupled, $K = 4$, $\lambda = 0.5$)	0.479	0.159

ground-truth shape and end-pose closely, while vision drifts laterally and IMU integrates a slowly-evolving pose offset.

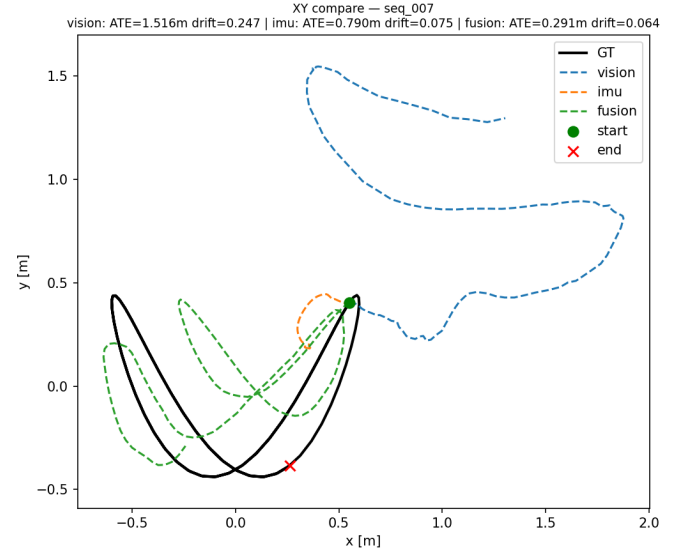


Fig. 6: Dead-reckoned trajectories on test sequence seq_007 on xy-plane. GT (black), vision (blue), IMU (orange) and fusion (green). Fusion follows the GT closely and ends near the GT terminal pose; vision drifts substantially away. Fusion ATE 0.291 m, drift **0.064**.

F. Ablation

TABLE IV: Mean test-set metrics for the fusion model under incremental additions to the design.

Configuration	ATE [m]	Drift
SmoothL1 + shared MLP head	1.72	0.27
+ geodesic $SO(3)$ loss	1.58	0.26
+ realistic trajectory speeds	0.92	0.38
+ decoupled output heads	0.48	0.20
+ $K = 4$ traj. loss ($\lambda = 1.0$)	0.52	0.18
+ retrain from scratch ($\lambda = 0.5$)	0.48	0.16

Decoupling the output heads delivers the largest single jump. The geodesic loss acts mostly on rotation accuracy. Capping the trajectory peak speed at ~ 2 m/s in the data generator (tightening the amplitude/period sampling ranges in Sec. II) primarily helps vision, whose ATE goes from

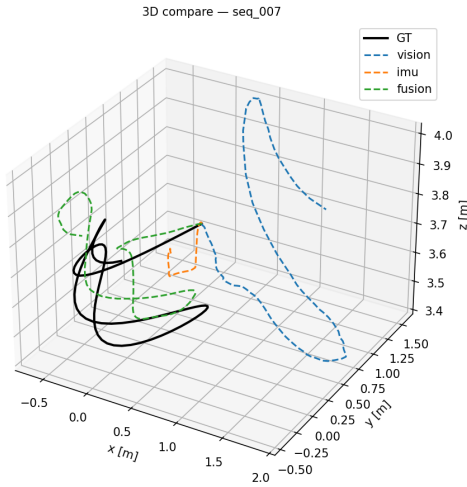


Fig. 7: Dead-reckoned trajectories on test sequence `seq_007` on xyz-plane. GT (black), vision (blue), IMU (orange) and fusion (green). Fusion follows the GT closely and ends near the GT terminal pose; vision drifts substantially away. Fusion ATE 0.291 m, drift **0.064**.

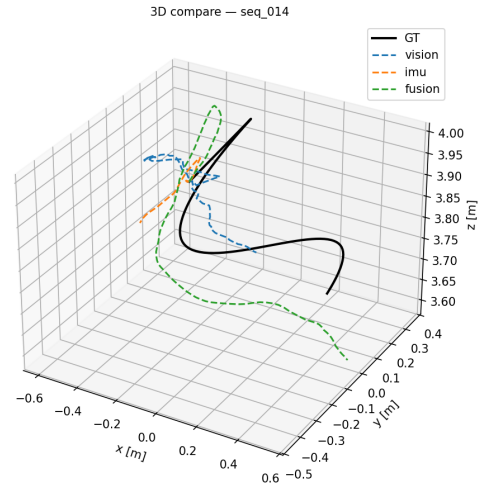


Fig. 9: Test sequence `seq_014` xyz-plane: an oscillatory sinusoidal trajectory where fusion correctly tracks the wave pattern with a slight magnitude undershoot. Fusion ATE 0.178 m, drift 0.120.

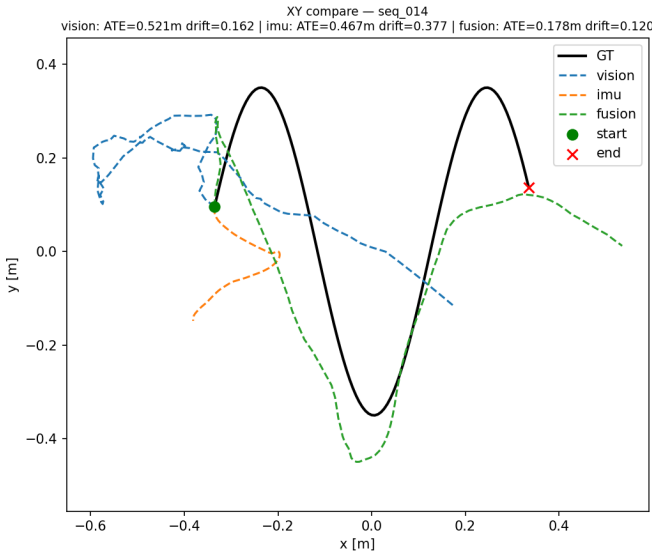


Fig. 8: Test sequence `seq_014` xy-plane: an oscillatory sinusoidal trajectory where fusion correctly tracks the wave pattern with a slight magnitude undershoot. Fusion ATE 0.178 m, drift 0.120.

≈ 2.3 m down to ≈ 0.79 m through the entire chain. The first $K = 4$ trajectory-loss attempt with $\lambda = 1.0$ achieved its intended drift reduction ($0.20 \rightarrow 0.18$) but at a small ATE cost ($0.48 \rightarrow 0.52$), because the trajectory term over-weighted endpoint consistency relative to pointwise accuracy. Re-training fusion from scratch with $\lambda = 0.5$ recovers the per-step loss weight while preserving the trajectory regularization, yielding the lowest ATE (0.48 m) and the lowest drift (0.16) of any configuration we tried.

VIII. CHALLENGES

a) *Regression to the dataset mean under SmoothL1-only training:* With the naive SmoothL1 loss and no trajectory term, all three networks shrank their predictions toward the training-set mean. The structural fixes including geodesic loss, decoupled heads, and K -step loss, eliminate this for fusion.

b) *IMU branch as a constant translator:* A 2-layer GRU on raw 6-DoF IMU samples cannot perform implicit double integration without an estimate of the initial velocity. The network solves the problem by emitting $\hat{t} \approx 0$, correct in the L1 sense, useless in practice. The fusion architecture sidesteps this by giving the rotation head IMU access and the translation head vision access.

c) *Closed-loop test sequences inflate the drift-ratio metric:* When the GT path nearly returns to its starting point, the path-length denominator stays large but the final-error numerator can be larger than the absolute path-traversed distance, producing apparently huge ratios. We report drift ratios as-is and use ATE as the primary metric.

IX. DISCUSSION

The fusion model halves both unimodal baselines on dead-reckoned trajectory metrics, despite being conceptually simple. Three structural choices were responsible for this result: (i) decoupled output heads that route rotation through IMU and translation through vision-and-IMU; (ii) the geodesic $SO(3)$ rotation loss, which respects the manifold structure of the rotation target; and (iii) the K -step trajectory composition loss, which trains the network on the same sequential composition operation used during test-time dead reckoning. The per-axis variance analysis supports this interpretation: IMU alone explains rotation almost perfectly but collapses toward a near-constant translation prediction, whereas vision provides the translation signal that the fusion head can exploit.

The main limitation of the final fusion model is its validation behavior. Although the best-validation checkpoint produces the strongest test-set trajectory results, the later epochs show a clear train-validation gap. This suggests that the fused representation has enough capacity to specialize to the finite set of synthetic training trajectories and textures after the early optimum. In the present work we address this operationally through checkpoint selection based on validation loss. As future work, this overfitting could be reduced more systematically by expanding the synthetic domain randomization, increasing texture and trajectory diversity, using stronger regularization in the fusion heads, applying modality dropout so that the network cannot over-rely on a single feature stream, and tuning early stopping or learning-rate schedules specifically for the fused architecture. Longer trajectory-composition windows may also reduce drift in nearly closed-loop sequences, but they should be introduced together with stronger regularization to avoid over-weighting endpoint consistency at the expense of pairwise accuracy.

X. RESEARCH DIRECTIONS

a) Classical VIO:

- More robust feature handling: better outlier rejection, adaptive feature selection, and stronger geometric consistency under motion blur, repetitive texture, or poor feature distribution.
- Online spatial and temporal calibration of camera-IMU extrinsics.
- Local filter-based VIO + global pose-graph optimization, pairing MSCKF-style efficient short-term estimation with long-term loop closure.
- Observability-aware filtering [4] to preserve the correct unobservable directions.
- Event-camera VIO: asynchronous brightness-change events deliver microsecond latency and high dynamic range, but require entirely new measurement models.

b) Deep VIO:

- Sequence-to-sequence IMU encoding with a wider temporal window so the IMU branch can infer velocity context instead of collapsing toward the dataset-mean translation.
- Stronger fusion regularization, including modality dropout, increased dropout in the fusion heads, weight-decay tuning, and early-stopping criteria selected specifically for the fused model.
- Broader synthetic domain randomization over textures, lighting, motion amplitudes, camera height, and IMU noise parameters to reduce the train-validation gap observed in the fusion loss curves.
- Differentiable Kalman or factor-graph layers wrapping the learned relative-pose head with a structured prior, restoring MSCKF-style consistency guarantees.
- Self-supervised photometric warp loss between consecutive frames, removing the dependence on synthetic GT and enabling fine-tuning on real flight footage.

- Bias and calibration estimation as auxiliary outputs, lifting the zero-initial-bias simplification.
- Geometry-aware backbones such as an explicit homography prediction head for planar scenes, with closed-form scale recovery (the route taken by PRGFlow [1] and CUAHN-VIO [2]).
- Sim-to-real domain adaptation against motion blur, rolling shutter, lighting changes, and real IMU bias drift.

XI. CONCLUSION

We designed and trained three deep VIO networks on a synthetic Blender dataset of an aerial robot flying over a textured floor. Geodesic $SO(3)$ rotation loss, output-head decoupling, and a K -step trajectory composition loss together bring the fusion model from a regime where it underperforms its vision-only baseline to one where it captures $\approx 88\%$ of per-step horizontal translation variance and essentially all of the rotation variance. On 15 unseen test sequences, the final fusion checkpoint selected by validation loss achieves mean dead-reckoned ATE ≈ 0.48 m and mean final-drift ratio ≈ 0.16 , nearly halving both unimodal baselines. The IMU-only results confirm that short-window inertial rotation is highly observable, while inertial translation remains ill-posed without velocity context; the fusion results show that this limitation can be mitigated by routing translation through the visual stream while preserving IMU-dominated rotation estimation. The largest remaining issues are integrated drift in challenging closed-loop sequences and mild overfitting of the fused model after its early validation optimum. These limitations motivate future work on longer sequence modeling, stronger fusion regularization, broader synthetic domain randomization, and structured differentiable VIO layers.

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